Axy are listed from 000 through 111 as shown in Fig. 5.17(b). The next-state values are obtained from the state equation

$$A(t + 1) = A \oplus x \oplus y$$

The expression specifies an odd function and is equal to 1 when only one variable is 1 or when all three variables are 1. This is indicated in the column for the next state of A.

The circuit has one flip-flop and two states. The state diagram consists of two circles, one for each state as shown in Fig. 5.17(c). The present state and the output can be either 0 or 1, as indicated by the number inside the circles. A slash on the directed lines is not needed, because there is no output from a combinational circuit. The two inputs can have four possible combinations for each state. Two input combinations during each state transition are separated by a comma to simplify the notation.

Practice Exercise 5.6

What determines the next state of a D-type flip-flop?

Answer: The next state of a D-type flip-flop is the value of D at the synchronizing edge of the clock.

Analysis with *JK* Flip-Flops

A state table consists of four sections: present state, inputs, next state, and outputs. The first two are obtained by listing all binary combinations. The output section is determined from the output equations. The next-state values are evaluated from the state equations. For a D-type flip-flop, the state equation is the same as the input equation. When a flip-flop other than the D type is used, such as JK or T, it is necessary to refer to the corresponding characteristic table or characteristic equation to obtain the next-state values. We will illustrate the procedure first by using the characteristic table and again by using the characteristic equation.

The next-state values of a sequential circuit that uses JK- or T-type flip-flops can be derived as follows:

1. Determine the flip-flop input equations in terms of the present state and input variables.
2. List the binary values of each input equation.
3. Use the corresponding flip-flop characteristic table to determine the next-state values in the state table.

As an example, consider the sequential circuit with two JK flip-flops A and B and one input x, as shown in Fig. 5.18. The circuit has no outputs; therefore, the state table does not need an output column. (The outputs of the flip-flops may be considered as the outputs in this case.) The circuit can be specified by the flip-flop input equations

$$J_A = B \quad K_A = Bx'$$
$$J_B = x' \quad K_B = A'x + Ax' = A \oplus x$$

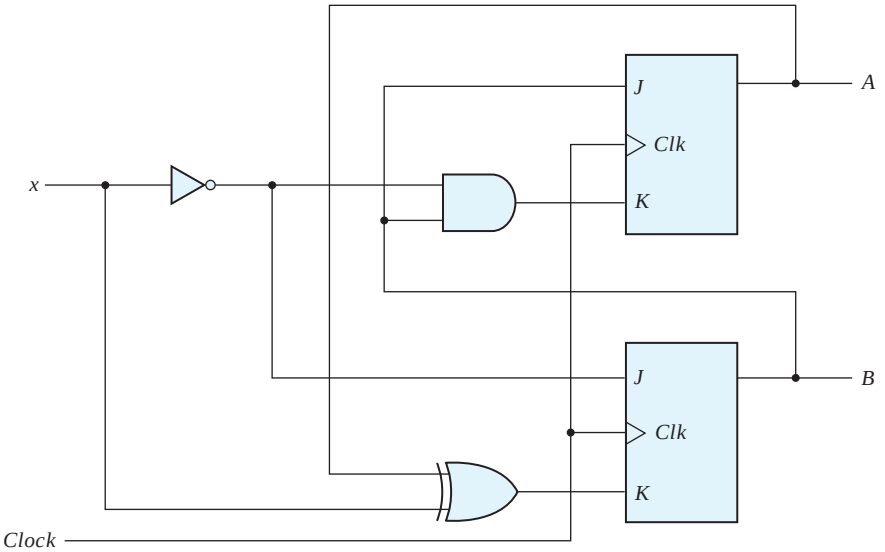


FIGURE 5.18
Sequential circuit with *JK* flip-flop

The state table of the sequential circuit is shown in Table 5.4. The present state and input columns list the eight binary combinations. The binary values listed under the columns labeled *flip-flop inputs* are not part of the state table, but they are needed for the purpose of evaluating the next state as specified in step 2 of the procedure. These binary values are obtained directly from the four input equations in a manner similar to that for obtaining a truth table from a Boolean expression. The next state of each flip-flop is evaluated from the corresponding J and K inputs and the characteristic table of the JK flip-flop listed in Table 5.1. There are four cases to consider. When $J = 1$ and $K = 0$, the

Table 5.4
State Table for Sequential Circuit with JK Flip-Flops

Present State		Input	Next State		Flip-Flop Inputs			
A	*B*	*x*	*A*	*B*	J_A	K_A	J_B	K_B
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	1	1	1	1	0
0	1	1	1	0	1	0	0	1
1	0	0	1	1	0	0	1	1
1	0	1	1	0	0	0	0	0
1	1	0	0	0	1	1	1	1
1	1	1	1	1	1	0	0	0

next state is 1. When $J = 0$ and $K = 1$, the next state is 0. When $J = K = 0$, there is no change of state and the next-state value is the same as that of the present state. When $J = K = 1$, the next-state bit is the complement of the present-state bit. Examples of the last two cases occur in the table when the present state AB is 10 and input x is 0. J_A and K_A are both equal to 0 and the present state of A is 1. Therefore, the next state of A remains the same and is equal to 1. In the same row of the table, J_B and K_B are both equal to 1. Since the present state of B is 0, the next state of B is complemented and changes to 1.

The next-state values can also be obtained by evaluating the state equations from the characteristic equation. This is done by using the following procedure:

1. Determine the flip-flop input equations in terms of the present state and input variables.
2. Substitute the input equations into the flip-flop characteristic equation to obtain the state equations.
3. Use the corresponding state equations to determine the next-state values in the state table.

The input equations for the two JK flip-flops of Fig. 5.18 were listed a couple of paragraphs ago. The characteristic equations for the flip-flops are obtained by substituting A or B for the name of the flip-flop, instead of Q:

$$A(t + 1) = JA' + K'A$$
$$B(t + 1) = JB' + K'B$$

Substituting the values of J_A and K_A from the input equations, we obtain the state equation for A:

$$A(t + 1) = BA' + (Bx')'A = A'B + AB' + Ax$$

The state equation provides the bit values for the column headed "Next State" for A in the state table. Similarly, the state equation for flip-flop B can be derived from the characteristic equation by substituting the values of J_B and K_B:

$$B(t + 1) = x'B' + (A \oplus x)'B = B'x' + ABx + A'Bx'$$

The state equation provides the bit values for the column headed "Next State" for B in the state table. Note that the columns in Table 5.4 headed "Flip-Flop Inputs" are not needed when state equations are used.

The state diagram of the sequential circuit is shown in Fig. 5.19. Note that since the circuit has no outputs, the directed lines out of the circles are marked with one binary number only, to designate the value of input x.

Practice Exercise 5.7

What determines the next state of a JK-type flip-flop?

Answer: The next state of a JK-type flip-flop is determined by the value of inputs J and K at the synchronizing edge of the clock.

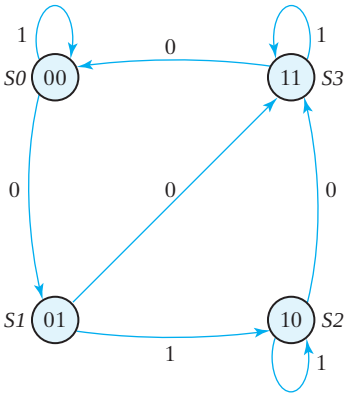


FIGURE 5.19
State diagram of the circuit of Fig. 5.18

Analysis with *T* Flip-Flops

The analysis of a sequential circuit with T flip-flops follows the same procedure outlined for JK flip-flops. The next-state values in the state table can be obtained by using either the characteristic table listed in Table 5.1 or the characteristic equation

$$Q(t + 1) = T \oplus Q = T'Q + TQ'$$

Now consider the sequential circuit shown in Fig. 5.20. It has two flip-flops A and B, one input x, and one output y and can be described algebraically by two input equations and an output equation:

$$T_A = Bx$$
$$T_B = x$$
$$y = AB$$

The state table for the circuit is listed in Table 5.5. The values for y are obtained from the output equation. The values for the next state can be derived from the state equations by substituting T_A and T_B in the characteristic equations, yielding

$$A(t + 1) = (Bx)'A + (Bx)A' = AB' + Ax' + A'Bx$$
$$B(t + 1) = x \oplus B$$

The next-state values for A and B in the state table are obtained from the expressions of the two state equations.

The state diagram of the circuit is shown in Fig. 5.20(b). As long as input x is equal to 1, the circuit behaves as a binary counter with a sequence of states 00, 01, 10, 11, and back to 00. When $x = 0$, the circuit remains in the same state. Output y is equal to 1 when the present state is 11. Here, the output depends on the present state only and is independent of the input. The two values inside each circle and separated by a slash are for the present state and output.